Over the past 75 years the study of kaon physics has been integral to the development of the standard model. Since 2016, the NA62 experiment has been the leading experiment in the field of kaon physics. However, the future of kaon physics is uncertain , with the NA62 experiment funded up to CERN LS3, which is expected at the end of 2025. It was then required to investigate the future of kaon physics and the potential experiments that could succeed NA62. This chapter will discuss the potential future of kaon physics, the experiments that could be used to further the field, and the work contributed to these experiments.

0.1 High Intensity Kaon Experiments (HIKE) Description

The High-intensity kaon experiments (HIKE) was a proposed project to continue kaon physics at CERN in the ECN3 experimental hall and represented a long-phased plan after LS3. HIKE would of profited from a four to six-fold increase in beam intensity compared to NA62 and included cutting-edge detector technologies that will be desicred in this section. There were three phases within the HIKE program, the first being a continuation of NA62's charged kaon physics with the high-intensity beam and detector upgrades. The second would have involved neural kaon physics with charged secondary particle tracking, and the third phase would have been designed to measure the $K_L \to \pi^0 \nu \bar{\nu}$ decay.

0.1.1 Phase 1

The experimental design of NA62 has been proven as a successful setup for the precision measurement of the $K^+ \to \pi^+ \nu \bar{\nu}$ decay. The HIKE phase 1 detector setup is shown in figure 1 with new and upgraded detectors with respect to the NA62 setup described in Section ??. The new and upgraded detectors are required due to the four times greater particle rate with respect to NA62 equating to 1.2×10^{13} protons per spill on the target. Other than the intensity upgrade, the beam had similar characteristics to NA62 with a $75\,Gev/c$ secondary beam produced from the $400\,GeV/c$ primary beam from the SPS accelerator onto a target at an angle of $0\,rad$.

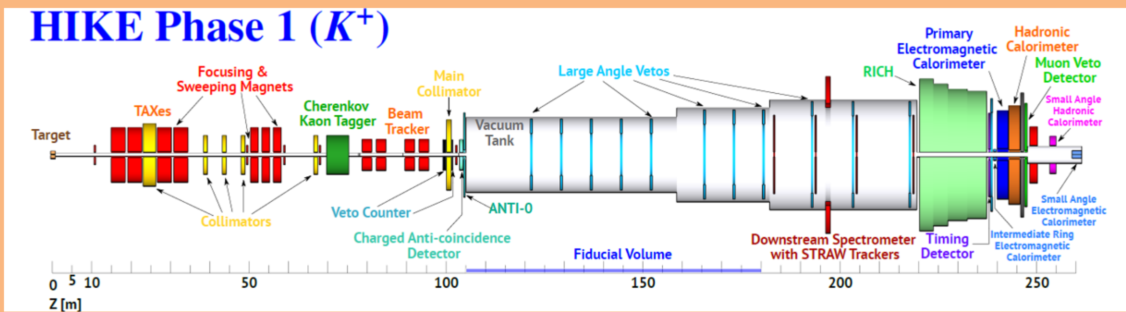


Figure 1: Diagram of HIKE phase 1 detector setup [1]

The upgraded detectors before the fiducial volume included an upgraded photon-detection system for the KTAG, described in section 0.2 and a new GTk chip design described in section 0.3. It was proposed to upgrade the existing NA62 Vetocounter with scintillating fiber (SciFi) technologies to improve the granularity and to accommodate the increased beam intensity.

Major upgrades were envisioned to the LAV and STRAW detectors within the fiducial volume. Some of the LAV stations would have been upgraded to a new design proposed for a lead scintillator design, similar to the Vacuum Veto System (VVS) planned to be used at the CKM experiment. The LAV detectors were important for the next phases of HIKE, especially for Phase 3 and would have been designed for the requirements of the $K_L \to \pi^0 \nu \bar{\nu}$ measurement even when used in this Phase. The STRAW spectrometer would have been upgraded to use a new 5 mm straw design shown in Figure 2. The reduction in straw diameter would have decreased drift times and increase granularity resulting in an expected improvement in the track angle and momentum resolution of 10-20 % compared to NA62.

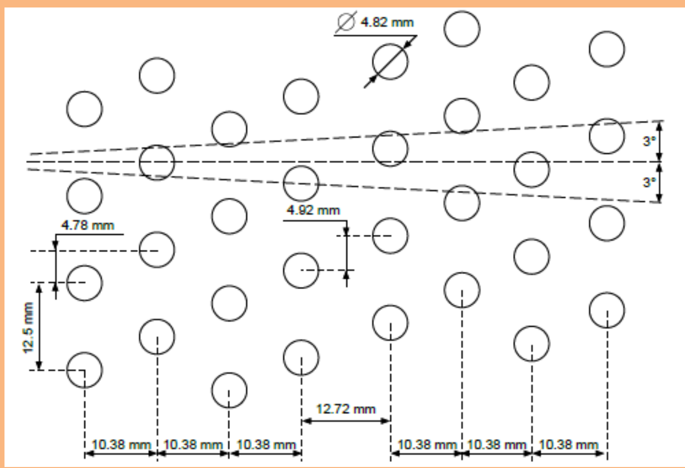


Figure 2: Diagram of the expected layout of the 5 mm straws [1]

Downstream of the fiducial volume, the PMT arrays of the RICH detector would have been upgraded, which is described in Section 0.4. The timing detector would profit from the current NA62-CHOD detector with upgraded PMTs, which would have been chosen from the number of studies from other detectors such as KTAG, RICH and MUV. For Phase 1, the primary electromagnetic calorimeter would have used the LKr from NA62, which had excellent energy resolution but was not satisfactory timing. The required timing resolution would have been achieved by upgrading the readout system. A number of other detectors would have been upgraded or replaced with higher granularity, such as the hadronic calorimeter, MUVs, IRC, and

SAC.

0.1.2 Phase 2

HIKE phase 2 would have been a multi-purpose neutral K_L decay experiment with the primary goal to measure the branching ratios of the $K_L \to \pi^0 l^+ l^-$ decays. The detector setup for phase 2, shown in Figure 3 would have involved minimal modification to the experimental setup of Phase 1, with the removal of detectors that were only applicable to the charge kaon beam. The KTAG, GTk, RICH, and SAC detectors would have been removed the straw spectrometer shortened to account for the softer momentum of the secondary particles, and the chambers would have been reconfigured to have central beam holes to account for the neural beam. Due to the compact detector, the fiducial volume would have been increased to 90 m compared to 75 m in Phase 1.

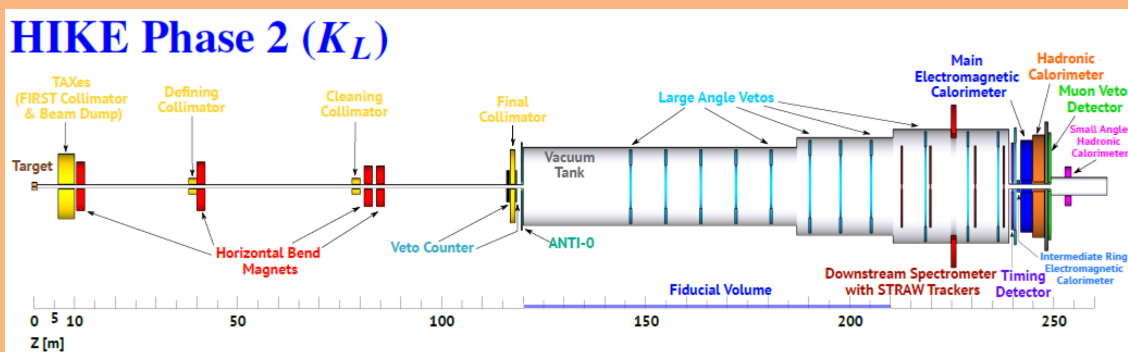


Figure 3: Diagram of HIKE phase 2 detector setup [1]

The design proposed a production angle of 2.4 mrad on the target producing an expected K_L yield of 5.4×10^{-5} per proton. Assuming an integrated proton flux of 1.2×10^{19} pot per year would have resulted in 3.8×10^{13} K_L decays in the fiducial volume per year. The mean momentum at the entry of the fiducial volume was expected to be 79 GeV/c with a mean momentum for K_L which decay of 46 GeV/c.

0.1.3 Phase 3 (KLEVER)

HIKE phase 3, previously known as KLEVER [2], was an experiment designed to measure the $K_L \to \pi^0 \nu \bar{\nu}$ decay. The experimental setup is shown in Figure 4 and differs from the design used at NA62 and HIKE Phase1/2. The setup consists of a large number of photon-detectors, located around a 160 m long vacuum volume to produce a complete coverage of photons produced from K_L decays. The photon-detectors include coverage for photons emitted up to 100 mrad and allow for a free path of photons to the main electromagnetic calorimeter (MEC) emitted up to 7.5 mrad.

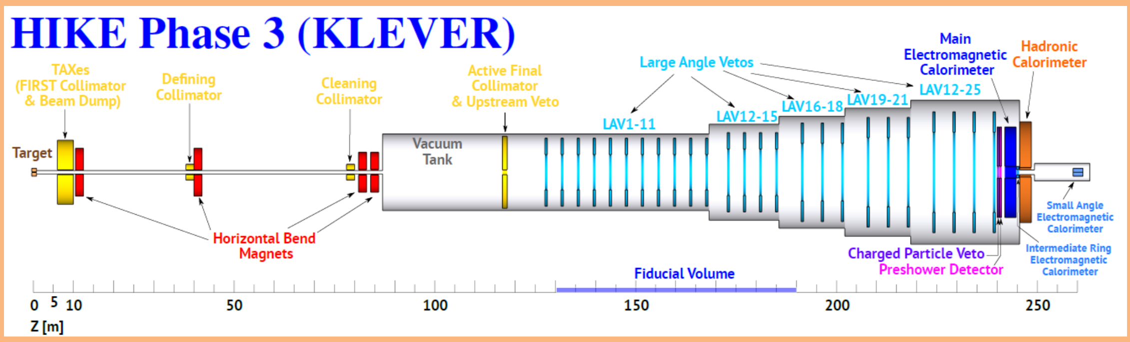


Figure 4: Diagram of HIKE phase 3 detector setup [1]

Due to the differing experimental setup, Phase 3 was separated from the HIKE proposal submitted in 2023. Still, some detectors would have been shared between the two experiments as the granularity and efficiency requirements for Phase 3 would have driven the detector requirements for the first two phases. The LAV's, MEC, IRC and SAC and the hadronic calorimters designed an built for HIKE Phase 1 and 2 would of been used in Phase 3. The upstream veto counter, charged particle veto, and the pre-shower detector are unique to Phase 3.

0.2 KTAG Photon Detection System upgrade

To sustain the increased kaon rate, the KTAG required upgrades to the photon-detection system and data acquisition system, with an expected time resolution of 15 - 20 ps and a kaon identification efficiency of greater than 95 %. The expected kaon rate was 200 MHz which corresponds to a maximum particle rate 10 MHz/cm^2 of photons at the PMT arrays, taking into the production of Cherenkov photons in CEDAR-H and the resultant acceptance of the photons. It was proposed that the PMT arrays would be replaced with arrays of micro-channel plate photomultipliers (MCP-PMTs), which provide a single photon time resolution of 50 - 70 ps depending on the model, a low dark rate of 1 kHz/cm^2 and the ability to sustain the required photon rate. A downside of the MCP-PMTs is their limited lifetime from the aging of the Photo-Cathode caused by heavy feedback ions from the residual gas. This results in the rapid degradation of PMT quantum efficiency with the increasing integrated anode charge.

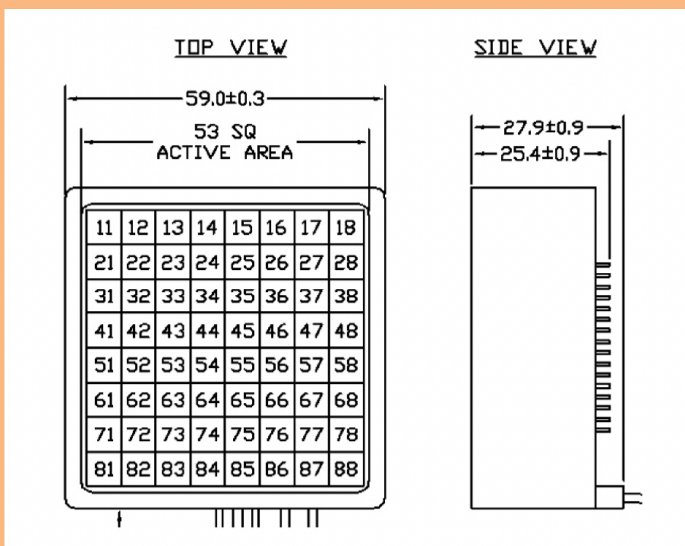


Figure 5: Scematic of Photonics MCP-PMT. CITE!

0.3 GTk upgrade design

0.4 RICH upgrade design

The overall design of the NA62 RICH described in section **??** was well suited for use within HIKE with an upgrade to the Cherenkov photon-detection system. The NA62 RICH photon detection system comprises two arrays on either side of the beam pipe, equipped with Hamamatsu R7400-U03 PMTs placed in a pattern described in Figure 6. The PMTs have a single hit time resolution of 240 ps and a quantum efficiency of 20 %. Due to the placement of the PMTs, the distance between the center of each PMT is 18 mm, which is constrained by the sensor size and the geometry of PMT itself. This distance between sensors was the main contributor to the single hit position resolution of 4.7 mm, resulting in an overall ring radius resolution of 1.5 mm. The number of hits per ring candidate and the single hit time resolution produced a time resolution of 80 ps. Due to the increased beam intensity, the main requirement of the RICH for HIKE was a time resolution of 20 - 30 ps.

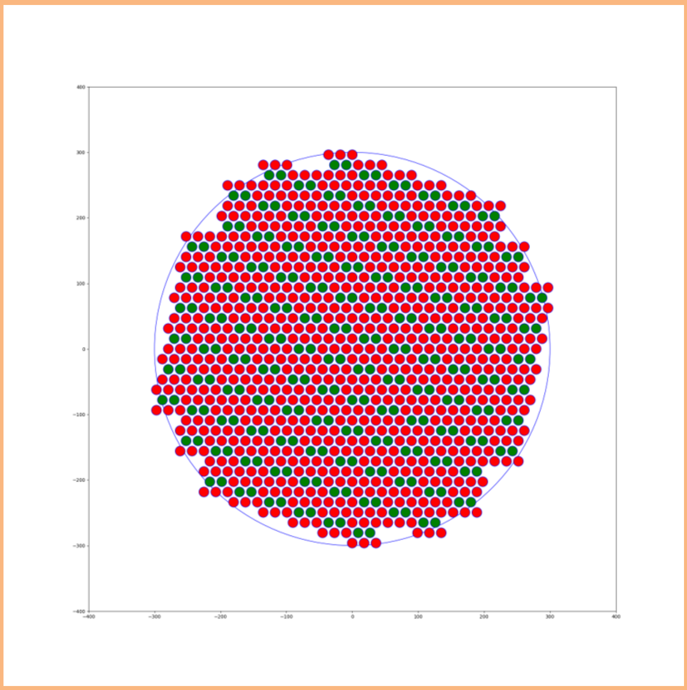


Figure 6: Diagram of RICH PMT array

The proposed RICH photon detection system upgrade would replace the current PMTs with a smaller sensor size, allowing for a smaller distance between sensor centers and reducing the hit position resolution. An excellent candidate for a sensor replacement was silicon photomultipliers (SiPMs), with small sensor sizes, high quantum efficiency of 40 % and a single hit time resolution of at least 100 ps were available from major manufacturers. Three sensor sizes were selected for further investigation, $9 \times 9\,mm$, $6 \times 6\,mm$, and $3 \times 3\,mm$. Each size had apparent benefits and downsides, but the smaller sensors would produce a better hit position resolution but at an increased cost and complexity. A study was conducted to determine the optimal sensor size for the SiPMs, with the overall detector performance balanced with the cost and complexity.

Using the upgraded NA62 software package described in section **??**, it was possible to implement the new PMTs into the RICH Monte Carlo simulation with the implemented arrays shown in Figure 7. Compared to the arrays currently used in NA62, the PMTs have been aligned on the vertical axis, and because of the geometry of the new PMTs, the are of the light spot, shown in Figure 7 as a grey circle, is completely covered by the PMTs with no gaps between the sensors.

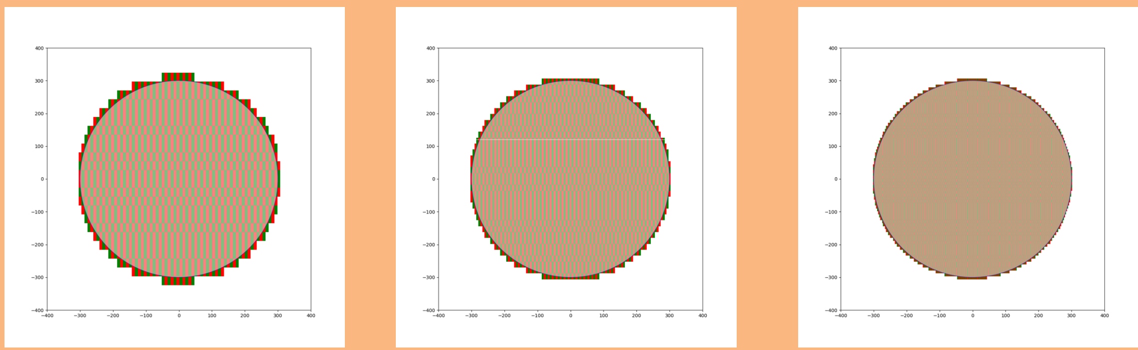


Figure 7

The letter of intent for HIKE proposed a SiPM with a Quantum efficiency of at least 40 % [3]. The Hamamatsu S13360-**50CS series was selected as the preliminary candidate for the RICH PMT upgrade to achieve this requirement. Hamamatsu offers a number of SiPMs in the S13360 series, with the **50CS selected for its improved Ultra Violet (UV) sensitivity and a high quantum efficiency of 40 % peak at 460 nm, shown in Figure 8.

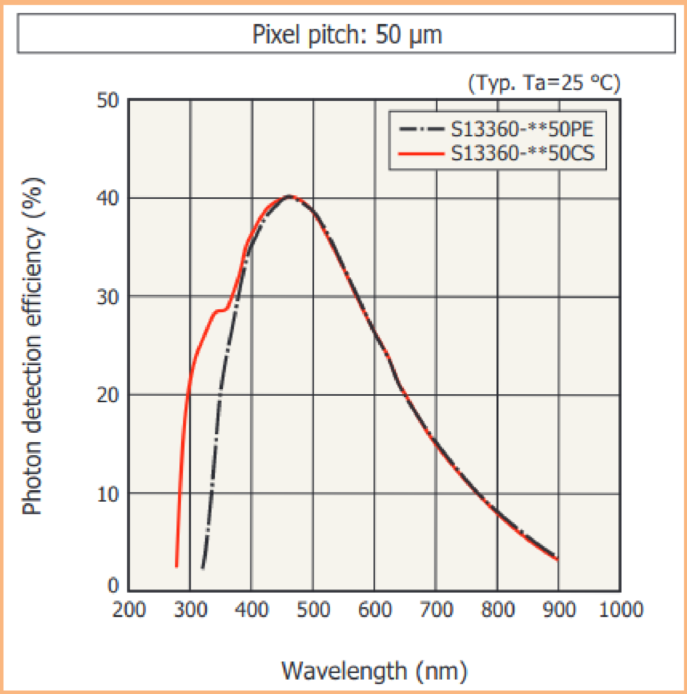


Figure 8: Selected SiPM quantum efficiency taken from Hamamatsu S13360 datasheet [4]

Along with the QE efficiency requirement was a single hit time resolution of 100 ps [3]. The manufacturer stated the Hamamatsu S13360-**50CS series has excellent time resolution, but no definite value was given. The time resolution would also depend on the final design of the RICH readout electronics and would have to be measured in a test setup. A simple Gaussian time response curve with a sigma of 100 ps was used for the simulations.

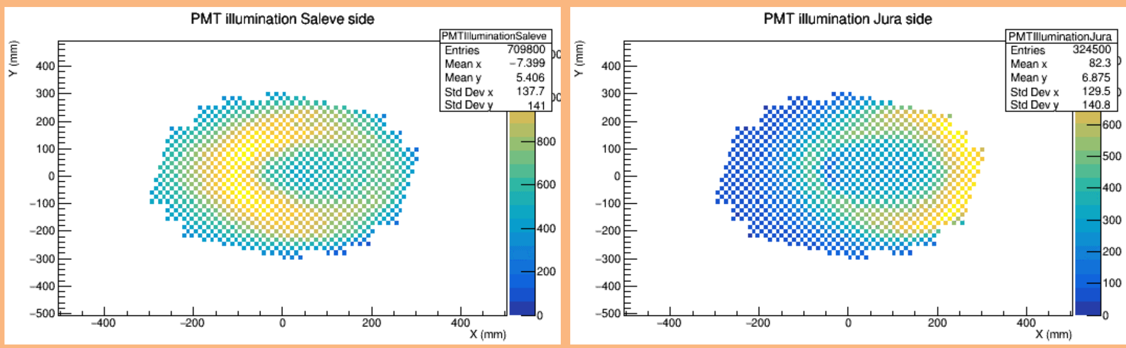
The expected performance of the RICH detector with the new PMTs was compared to the current NA62 RICH detector in Table 1. With the increased quantum efficiency, the expected number hits per π^+ at 15 and 45 GeV/c increased by a factor of two. This, along with the improved time resolution of the new PMTs resulted in a time π^+ time resolution of 27 ps at 15 GeV/c and 20 ps at 45 GeV/c. Using the implemented upgrades to the RICH detectors, the above expectations were confirmed

	NA62 RICH	HIKE RICH
Sensor time resolution	240ps	100ps
Sensor quantum efficiency	20%	40%
Number of hit for π^+ at 15 GeV/c	7	14
Number of hit for π^+ at 45 GeV/c	12	24
Time resolution for π^+ at 15 GeV/c	90ps	27ps
Time resolution for π^+ at 45 GeV/c	70ps	20ps

Table 1: Comparisons of the expected performance of the NA62 and HIKE RICH detectors [1]

by a Monte Carlo sample of $K^+ \to \pi^0 e^+ \nu$ (Ke3).

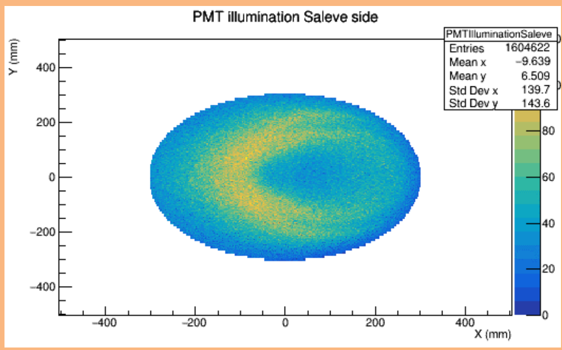
For the three sensor sizes, a test production of 1M Ke3 was used to determine the preliminary performance of the detector and compare this to the current NA62 setup. The plots in Figure 10 show the light distribution on the PMT arrays for the three sensor sizes: 3 mm 10(a,b), 6 mm 10(c,d), and 9 mm 10(e,f) compared to the illumination on the current NA62 PMT arrays 9. Due to the better placement of the square sensors, it is possible to see a more uniform illumination of the PMT arrays and better coverage of the light ring, especially in the Jura side. It is clear to see that either sensor size would be an improvement on the current NA62 setup.



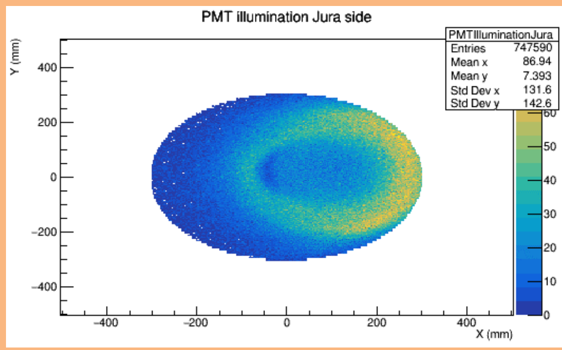
(a) Light distribution for NA62 setup on the Salev side of the RICH detector.

(b) Light distribution for NA62 setup on the Jura side of the RICH detector.

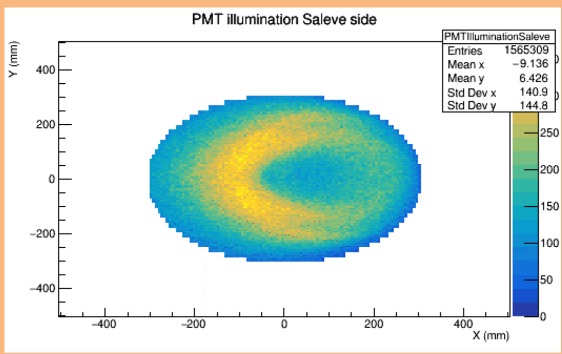
Figure 9: Cherenkov light illumination of RICH PMT arrays for the NA62 PMT arrays. The left column shows illumination on the Salev side of the RICH detector, and the right column shows the Jura side.



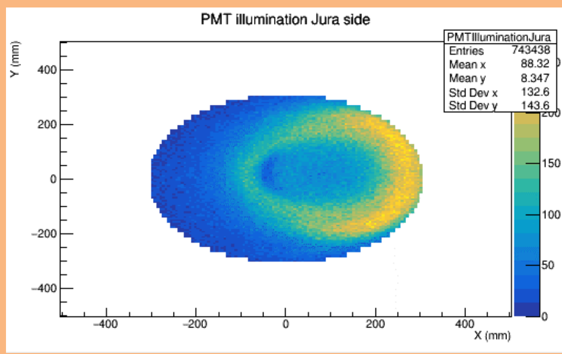
(a) Light distribution for 3 mm sensors on the Salev side of the RICH detector.



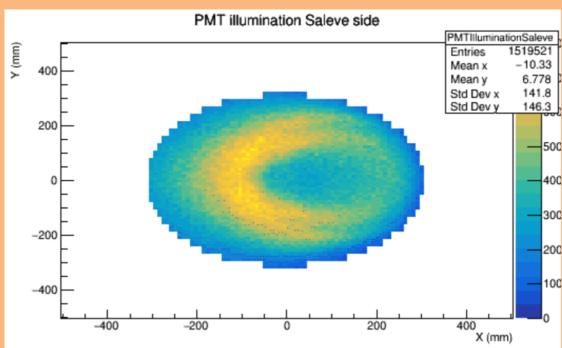
(b) Light distribution for 3 mm sensors on the Jura side of the RICH detector.



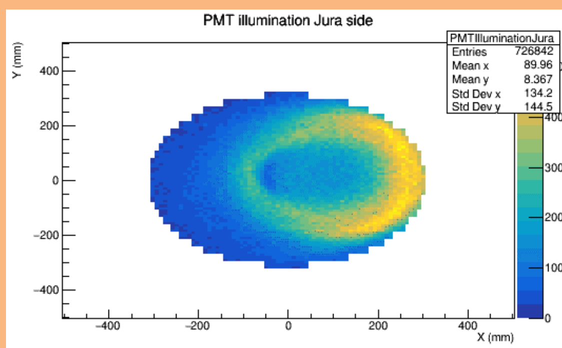
(c) Light distribution for 6 mm sensors on the Salev side of the RICH detector.



(d) Light distribution for 6 mm sensors on the Jura side of the RICH detector.

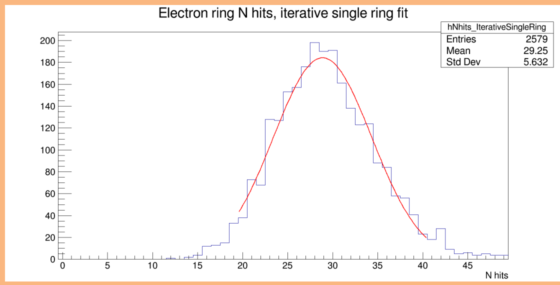


(e) Light distribution for 9 mm sensors on the Salev side of the RICH detector.



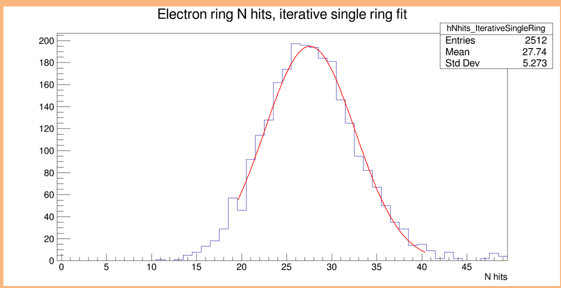
(f) Light distribution for 9 mm sensors on the Jura side of the RICH detector.

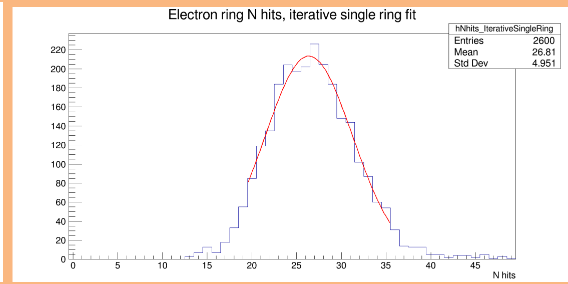
Figure 10: Cherenkov light illumination of RICH PMT arrays for the three different sensor sizes. The left column shows illumination on the Salev side of the RICH detector, and the right column shows the Jura side.



(a) Number of hits per ring candidate for NA62 PMT arrays.

(b) Number of hits per ring candidate for 3 mm SiPMs, = 28.86.

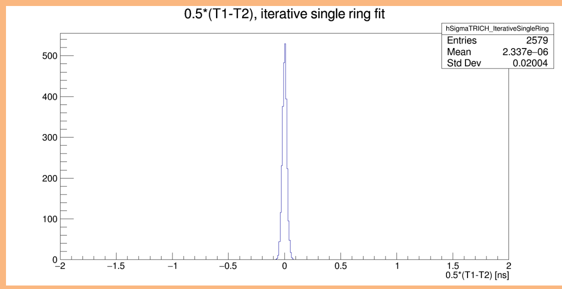




(a) Number of hits per ring candidate for 6 mm SiPMs = 27.56.

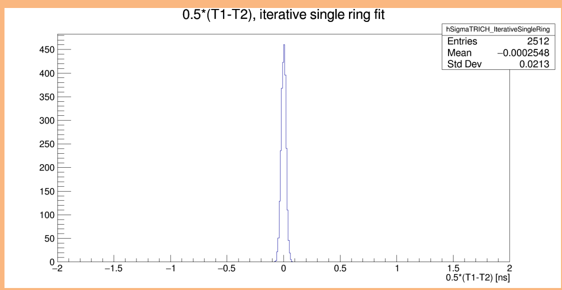
(b) Number of hits per ring candidate for 9 mm SiPMs = 26.36.

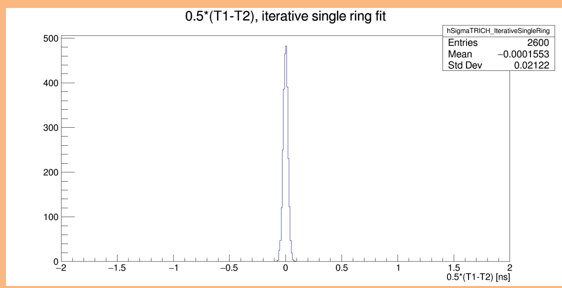
Figure 11: Number of hits per ring candidate for NA62 PMTs and the three possible HIKE SiPMs



(a) Time resolution for NA62 PMT arrays.

(b) Time resolution for 3 mm SiPMs.





(a) Time resolution for 6 mm SiPMs.

(b) Time resolution for 9 mm SiPMs.

Figure 12: Time resolution for NA62 PMTs and the three possible HIKE SiPMs, calculated from the time difference between two equally populated subset of hits within the ring candidate. A detailed description of the procedure is provided here [5]

0.5 Detector Rates

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